



INDIA.RUNS

Build what next India runs on

Ranking 100,000 Candidates The Way A Great Recruiter Would –
By Reading Career Evidence, Not Keywords.

Team

InnoCoders

Team Leader

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Problem Statement

**Candidate Ranking – Senior AI/ML
Engineer**

■ Solution Overview

02

A Recruiter-Grade Ranking Engine That Judges Career Evidence – Not Keywords.

100K**Candidates Scored**

Complete Pool, Single Pass

31s**End-To-End Runtime**

On One CPU Core

0**External Dependencies**

Pure Python Standard Library

Five-Dimension Scoring

Honeypot Defense

Career-Evidence Cross-Validation

Fully Explainable

■ JD Understanding & Candidate Evaluation

03

Target Role – Retrieval, Ranking & Recommendation Systems (Senior AI/ML).

Signals That Most Determine Fit

- 1 Career-Description Evidence**
The Single Hardest Signal To Fake
- 2 Role & Title Trajectory**
Current Title + Peak, Recency-Weighted
- 3 Production-Deployment Proof**
Shipped Systems, Not Just Research
- 4 Company Quality & Tenure**
Product Companies, Stability Over Hopping
- 5 Availability & Responsiveness**
Open-To-Work, Notice, Response Rate

Stated Disqualifiers

- Non-Technical Roles
- Consulting-Only Careers
- Research-Only (No Production)
- CV / Speech-Only (No NLP Crossover)
- Ghost Candidates

Ranking Methodology

04

Five Weighted Dimensions



● Technical Depth ● Title Relevance ● Career Quality ● Experience Fit ● Availability

Final Score

Base × **Location Multiplier** × **Disqualifier Penalties**

Honeypot Override

A Flagged Profile Is Forced To A Near-Zero Score (×0.02), Regardless Of The Formula Above.

Self-Reported Skills Are Cross-Validated Against Career Text · Deterministic Tie-Break By candidate_id.

■ Explainability & Data Validation

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Explainability

- Per-Candidate Reasoning Built Only From Facts In That Profile.
- Honest Concerns Surfaced – Notice Period, Response Rate, Location.
- Phrasing Varies By Rank Band; Never Templated.
- Zero Hallucinated Skills – Every Line Regex-Verified Against The Profile.

Honeypot Defense – Every Profile Verified Against Itself



Date Arithmetic

Duration Vs. Own Start/End Dates



Expert, Zero Use

Expert Skills, 0 Months Logged



YOE > Career Span

Stated Years Exceed History



Tech Anachronism

Skill Predates The Technology

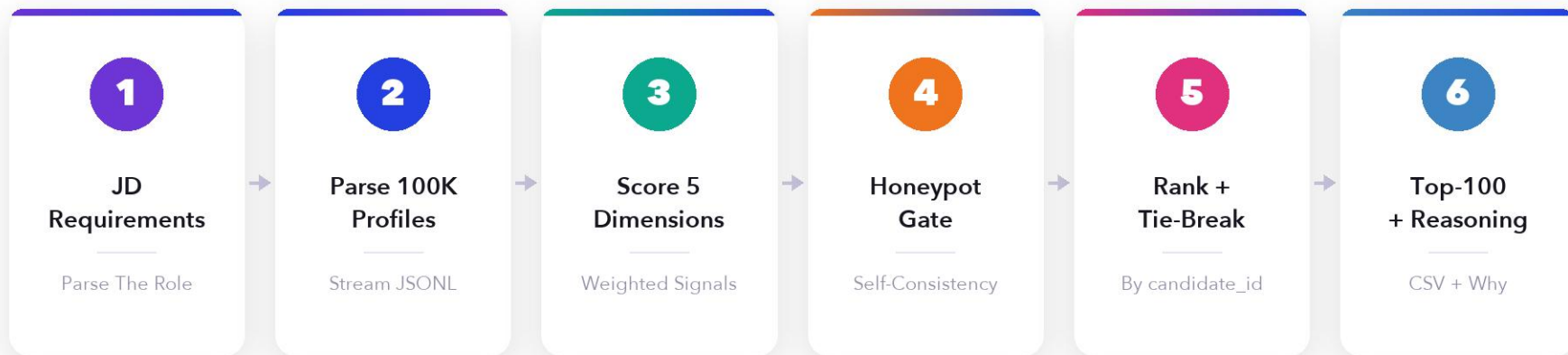
141**Impossible Profiles Caught**

0 In The Submitted Top-100 · Threshold Is 10%

■ End-To-End Workflow

06

Six Deterministic Stages – Input To Ranked Top-100 In A Single Pass.



Single Pass · ~31s On One CPU Core · Deterministic, Byte-Identical Reruns

■ System Architecture

07

Input

candidates.jsonl (100K Profiles) + JD-Derived Rule Configuration

Scoring Engine

Five Independent Standard-Library Scoring Modules

Title

Technical

Career

Experience

Availability

Validation

Keyword Cross-Check + honeypot_flags() Self-Consistency

Output

Validated Top-100 CSV + Per-Candidate Reasoning

Guarantees

- Deterministic
- Single Pass
- Stdlib-Only
- < 1 GB RAM
- CPU-Only
- ~31s

Results & Performance

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100,000**Candidates Scored**

Complete Pool

~31s**End-To-End Runtime**

One CPU Core

0**Dependencies**

Pure Standard Library

141**Honeypots Caught**

Pool-Wide Checks

0%**Honeypots In Top-100**

DQ Threshold 10%

< 1 GB**Peak Memory**

16 GB Budget

Constraints Met

**Runtime**
31s / 5min**CPU-Only**
Yes**No Network**
Yes**Deterministic**
Verified**Validator**
Passes

Technologies Used

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Python 3.9+ Standard Library Only

The Entire Ranking System, Built From Modules That Ship With Python.

json

csv

datetime

argparse

pathlib

re

Reproducible

Same Input, Byte-Identical Output

Portable

Runs Anywhere Python Runs; No Install

Auditable

Read The Entire Scoring Logic Line By Line

No GPU · No Network · No Model Downloads · Well Inside The Compute Budget

■ Submission Assets

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■ GitHub Repository

De-Coder05/redrob-ranking-challenge

■ Ranked Output

InnoCoders.csv · Top-100,
Validator-Clean

■ Approach Deck

approach_deck.pdf

■ Ranking Engine

rank.py · The Entire System,
Stdlib-Only

■ Submission Metadata

submission_metadata.yaml

■ Format Validator

validate_submission.py

Reproduce The Submission

```
$ python rank.py --candidates ./candidates.jsonl --out ./InnoCoders.csv
```



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THANK YOU